Python Engineering - Class 01 Reference Guide

From Equation to Algorithm

Reference guide to the syntax and tools used in this class. It explains what each element does, how it is written and when it is useful.

Quick map

Concept	Syntax	Idea
Variables	rho = 1.225	A name that refers to a value stored during execution.
Numeric types	int / float	Use int for whole numbers and float for engineering quantities with decimals.
Assignment	x = value	The right-hand side is evaluated first and assigned to the name on the left.
Arithmetic	+ - * / **	Python operators used to translate mathematical expressions.
Comparison	< <= > >=	Produces a Boolean value used for validation.
print()	print(value)	Displays values; f-strings allow labels, units and formatting.
f-string	f'{q:.2f}'	Insert values inside text and control decimal formatting.

1. Variables

Syntax: `rho = 1.225`

A name that refers to a value stored during execution.

```
rho = 1.225
speed = 100.0
```

Control question: before running the example, predict the result type and shape.

2. Numeric types

Syntax: `int / float`

Use int for whole numbers and float for engineering quantities with decimals.

```
n = 5          # int
rho = 1.225    # float
```

Control question: before running the example, predict the result type and shape.

3. Assignment

Syntax: `x = value`

The right-hand side is evaluated first and assigned to the name on the left.

```
q = 0.5 * rho * speed**2
```

Control question: before running the example, predict the result type and shape.

4. Arithmetic

Syntax: `+ - * / **`

Python operators used to translate mathematical expressions.

```
a = 2 + 3
b = 4 * 5
c = 10 / 2
d = 3**2
```

Control question: before running the example, predict the result type and shape.

5. Comparison

Syntax: `< <= > >=`

Produces a Boolean value used for validation.

```
valid = speed >= 0
```

Control question: before running the example, predict the result type and shape.

6. print()

Syntax: `print(value)`

Displays values; f-strings allow labels, units and formatting.

```
print(f'Dynamic pressure: {q:.2f} Pa')
```

Control question: before running the example, predict the result type and shape.

7. f-string

Syntax: `f'{q:.2f}'`

Insert values inside text and control decimal formatting.

```
print(f'q = {q:.2f} Pa')
```

Control question: before running the example, predict the result type and shape.

Common mistakes

- Confusing a function name with a function call using parentheses.
- Not checking types, indices or shapes before an operation.
- Using a tool because it works without being able to explain what it receives and returns.

Mental cheat sheet

```
Variables: rho = 1.225
Numeric types: int / float
Assignment: x = value
Arithmetic: + - * / **
Comparison: < <= > >=
print(): print(value)
f-string: f'{q:.2f}'
```